

```

67.
68.     # Draw the initial sand:
69.     for sand in allSand:
70.         bext.goto(sand[X], sand[Y])
71.         print(SAND, end='')
72.
73.     runHourglassSimulation(allSand)
74.
75.
76. def runHourglassSimulation(allSand):
77.     """Keep running the sand falling simulation until the sand stops
78.     moving."""
79.     while True: # Keep looping until sand has run out.
80.         random.shuffle(allSand) # Random order of grain simulation.
81.
82.         sandMovedOnThisStep = False
83.         for i, sand in enumerate(allSand):
84.             if sand[Y] == SCREEN_HEIGHT - 1:
85.                 # Sand is on the very bottom, so it won't move:
86.                 continue
87.
88.                 # If nothing is under this sand, move it down:
89.                 noSandBelow = (sand[X], sand[Y] + 1) not in allSand
90.                 noWallBelow = (sand[X], sand[Y] + 1) not in HOURGLASS
91.                 canFallDown = noSandBelow and noWallBelow
92.
93.                 if canFallDown:
94.                     # Draw the sand in its new position down one space:
95.                     bext.goto(sand[X], sand[Y])
96.                     print(' ', end='') # Clear the old position.
97.                     bext.goto(sand[X], sand[Y] + 1)
98.                     print(SAND, end='')
99.
100.                    # Set the sand in its new position down one space:
101.                    allSand[i] = (sand[X], sand[Y] + 1)
102.                    sandMovedOnThisStep = True
103.                else:
104.                    # Check if the sand can fall to the left:
105.                    belowLeft = (sand[X] - 1, sand[Y] + 1)
106.                    noSandBelowLeft = belowLeft not in allSand
107.                    noWallBelowLeft = belowLeft not in HOURGLASS
108.                    left = (sand[X] - 1, sand[Y])
109.                    noWallLeft = left not in HOURGLASS
110.                    notOnLeftEdge = sand[X] > 0
111.                    canFallLeft = (noSandBelowLeft and noWallBelowLeft
112.                    and noWallLeft and notOnLeftEdge)
113.
114.                    # Check if the sand can fall to the right:
115.                    belowRight = (sand[X] + 1, sand[Y] + 1)
116.                    noSandBelowRight = belowRight not in allSand
117.                    noWallBelowRight = belowRight not in HOURGLASS
118.                    right = (sand[X] + 1, sand[Y])
119.                    noWallRight = right not in HOURGLASS
120.                    notOnRightEdge = sand[X] < SCREEN_WIDTH - 1
121.                    canFallRight = (noSandBelowRight and noWallBelowRight

```